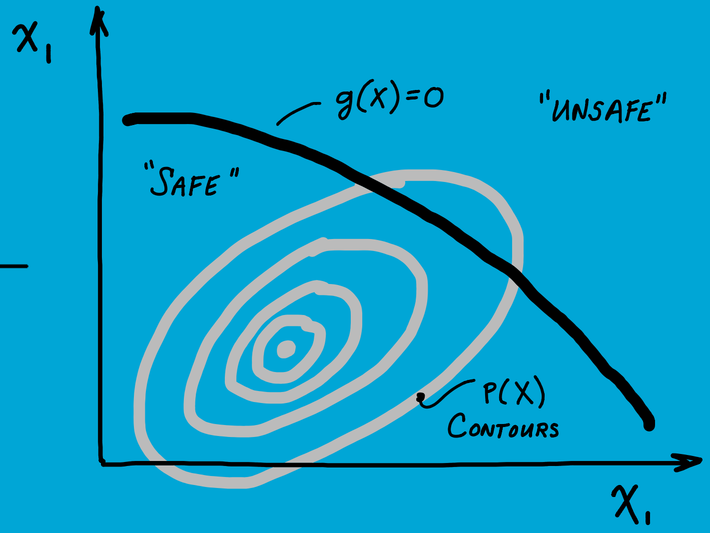
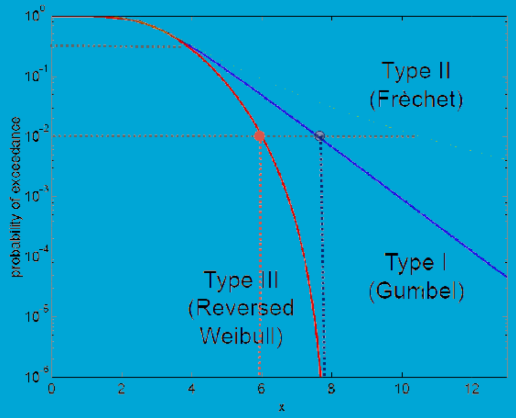
CIEM42X0 Probabilistic Design

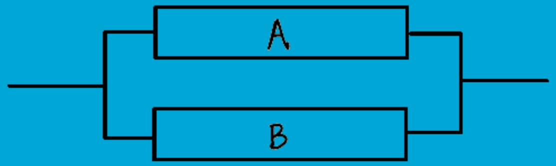
Hydraulic and Offshore Structures (HOS) Track

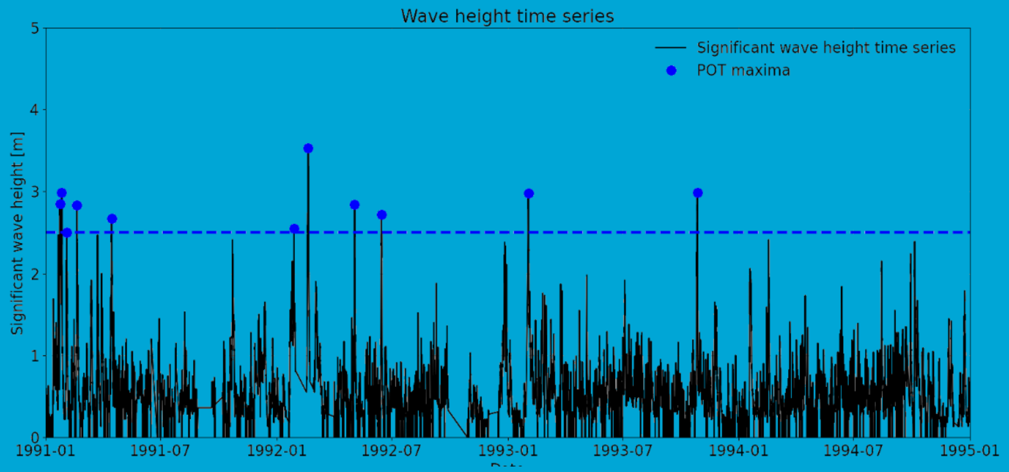
Civil Engineering MSc Program



Component reliability







Patricia Mares Nasarre

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Outline

1. **What is component reliability?**
2. Concept of failure in reliability
3. Method 1: Monte Carlo
4. Method 2: FORM

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1. What is component reliability?

Component reliability refers to the probability that a **single component** performs its intended function **without failure** for a specified period of time under stated conditions.

Two main perspectives:

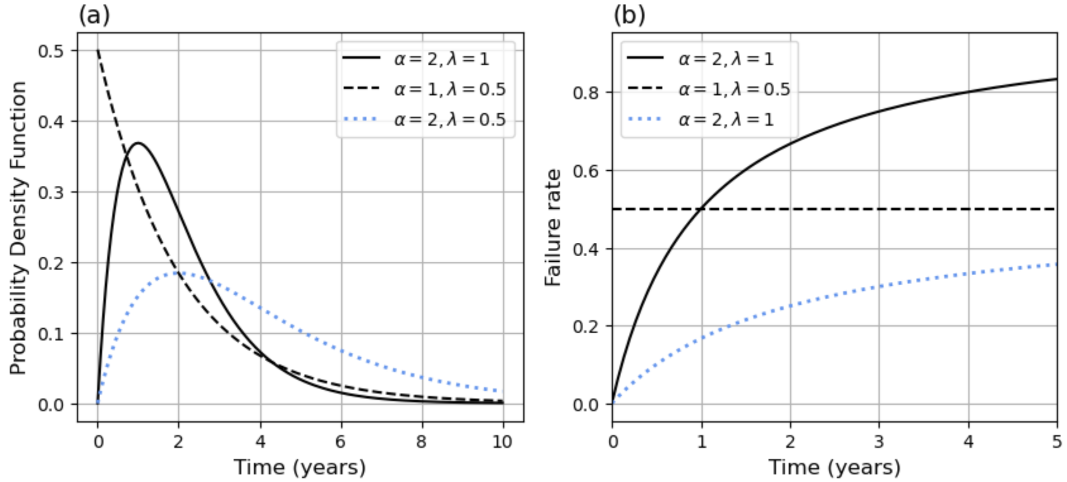
1. Failure rate (MUDE)

Failure distributions

A failure distribution is a model that describes mathematically the lifetime of a material, a devise or a structure or system. For instance, it may describe the uncertainty in the amount months before a light bulb burns out.

In this case, let X be a life variable. An important concept is the *failure rate* or *hazard rate* at a time t, denoted as $r_X(t)$ and given by

$$r_X(t) = \lim_{\Delta \to 0} \frac{P(t < X \leq t + \Delta t | X > t)}{\Delta t}$$



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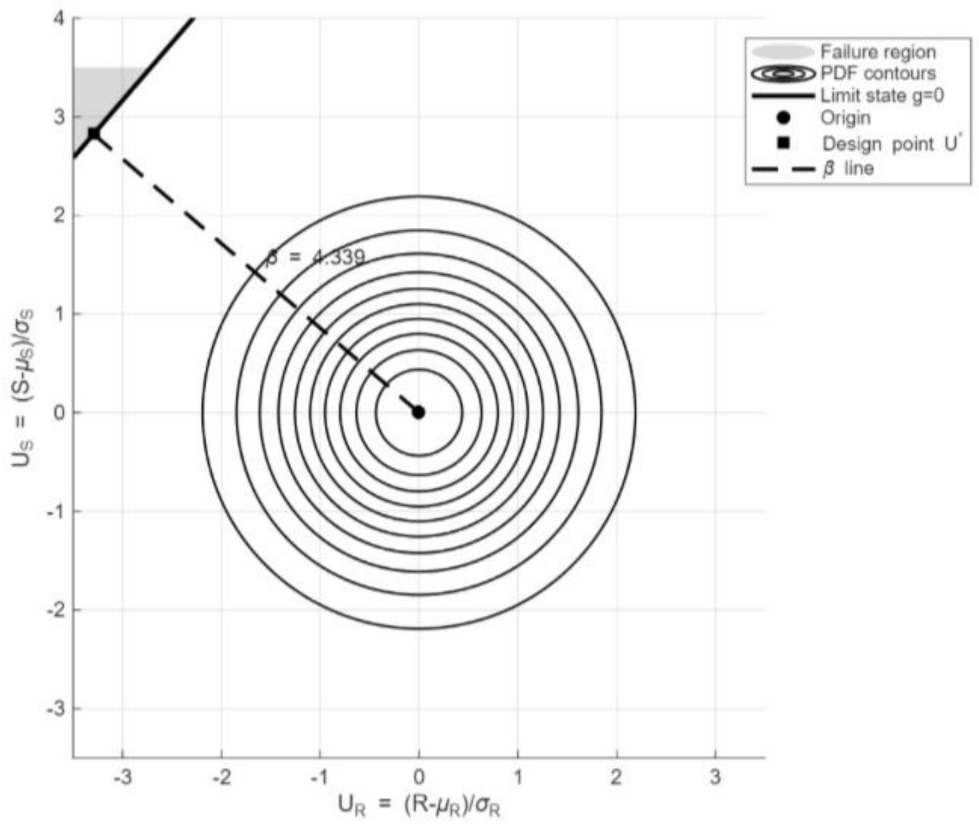
1. What is component reliability?

Component reliability refers to the probability that a **single component** performs its intended function **without failure** for a specified period of time under stated conditions.

Two main perspectives:

1. Failure rate (MUDE)

2. Limit state functions (MUDE & **this course**)



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Outline

1. What is component reliability?
2. **Concept of failure in reliability**
3. Method 1: Monte Carlo
4. Method 2: FORM

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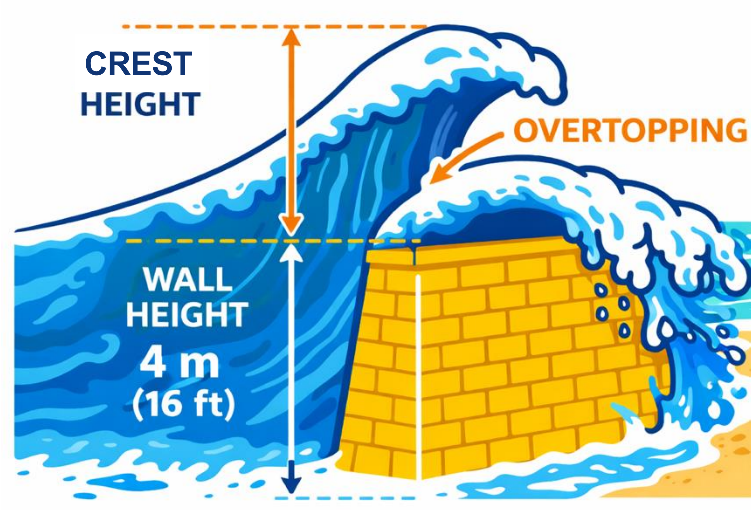
2. Concept of failure in reliability

We describe **failure** mathematically through a ***limit state function***, *g(x)*.

Typically, in the form of "**resistance - loading**", where the probability of failure is defined as

$$P_{failure} = \mathrm{P}[g(x) < 0]$$



If wave crest height > wall height → overtopping

Let's assume we cannot afford ANY overtopping. Thus,

Resistance =

Load =

Limit state function is:

Pf = P(h_c >h_w)

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2. Concept of failure in reliability

We describe **failure** mathematically through a ***limit state function***, *g(x)*.

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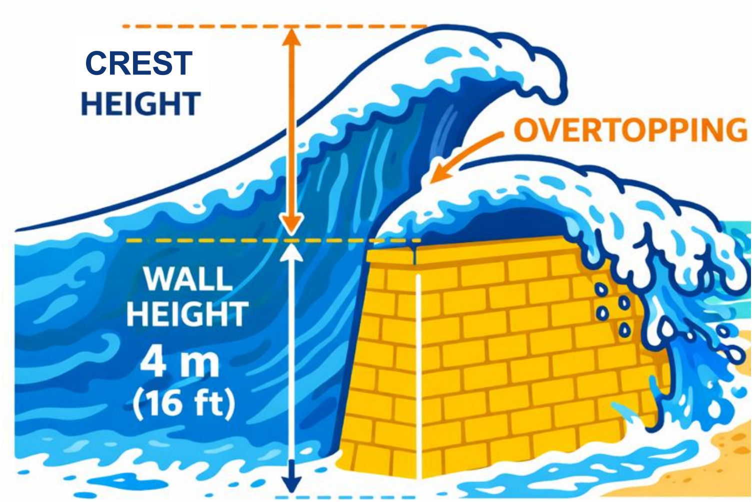


If wave crest height > wall height → overtopping

Let’s assume we cannot afford ANY overtopping. Thus,

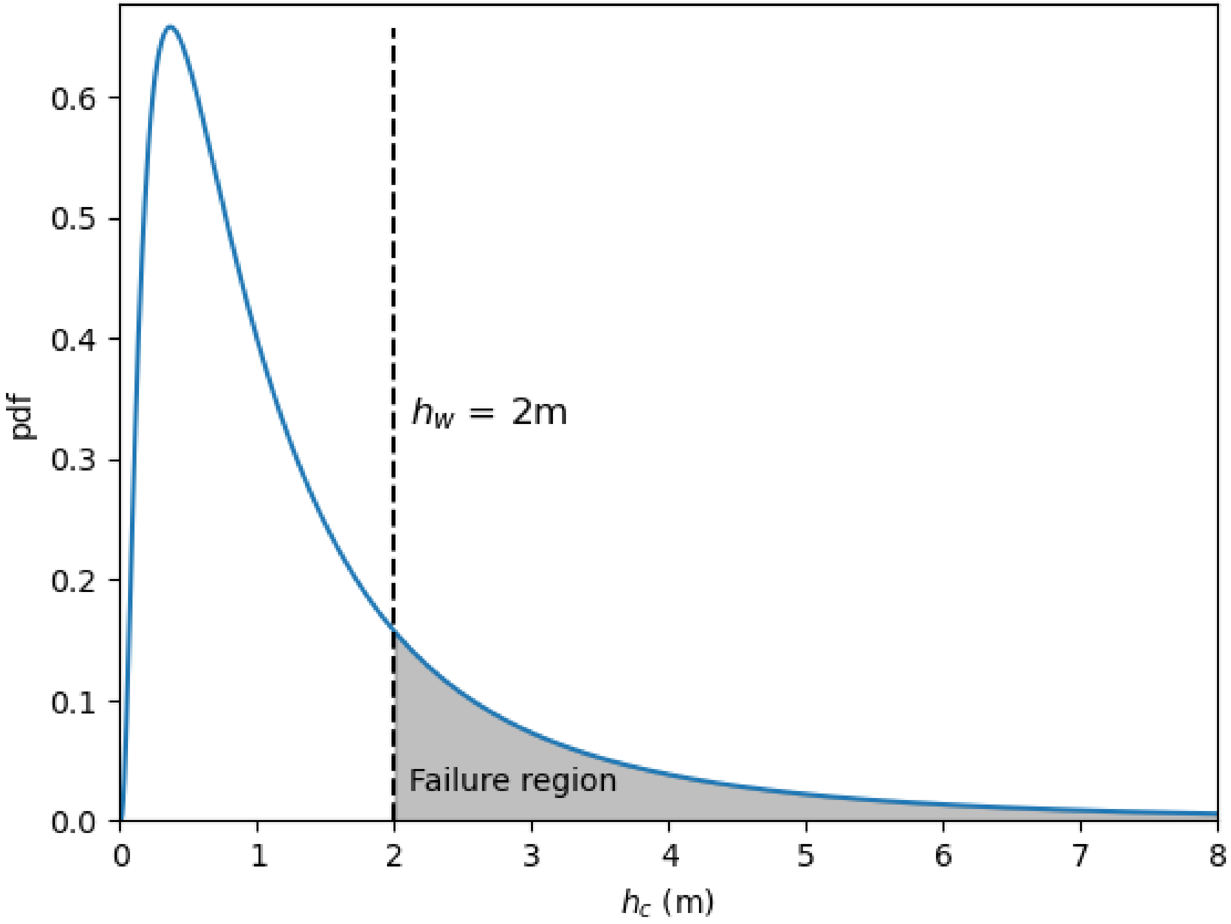
Resistance = height of the wall above MWL (h_w)

Load = wave crest height (h_c)

Limit state function is: $g(h_w, h_c) = h_w - h_c$

$Pf = P(h_c > h_w)$

2. Concept of failure in reliability



If wave crest height > wall height → overtopping

Let's assume we cannot afford ANY overtopping. Thus,

Resistance = height of the wall above MWL (h_w)

Load = wave crest height (h_c)

Limit state function is: $g(h_w, h_c) = h_w - h_c$

Pf = $P(h_c > h_w) = \int_{h_w}^{+\infty} f(h_c)\,dh_c$

2. Concept of failure in reliability

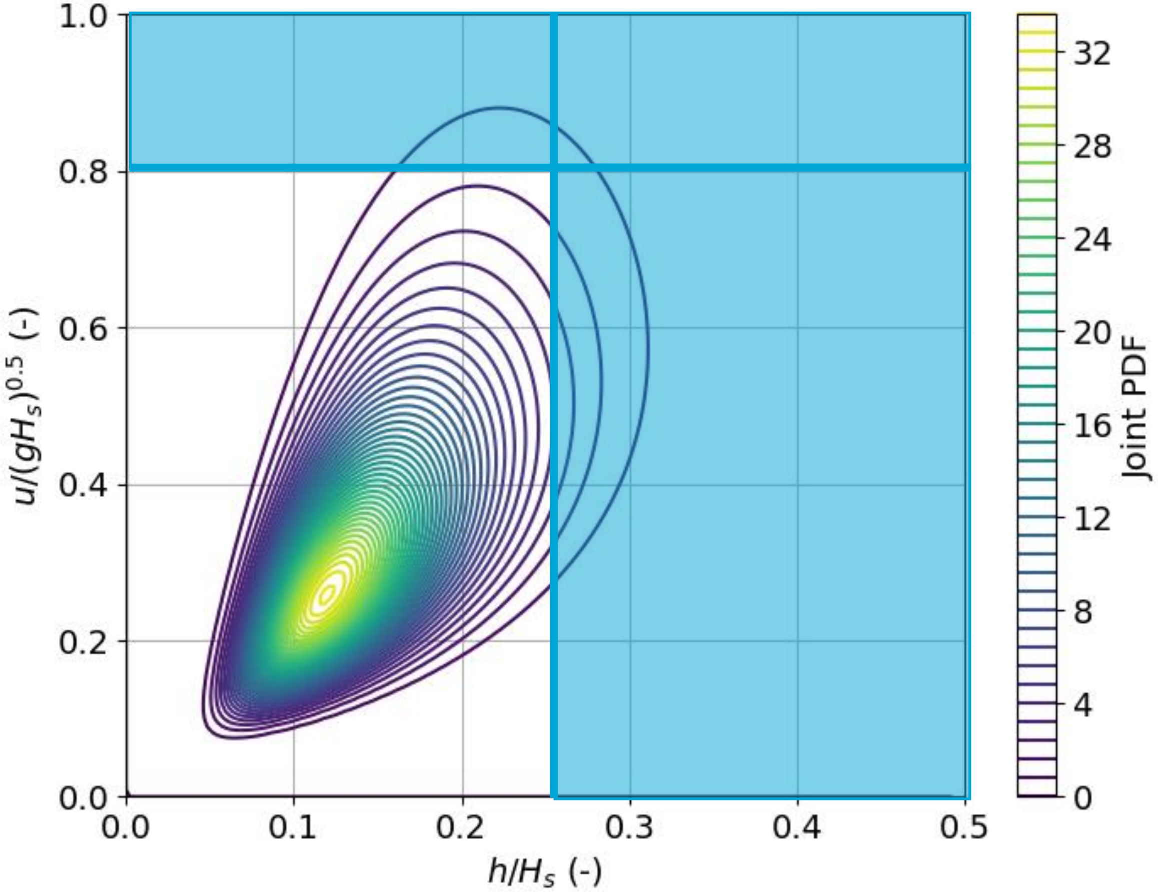
Problems are rarely univariate…

A more realistic example: wave overtopping a breakwater

- Operativity or pedestrian safety:
- ○ Max flow velocity on crest (here u)
- ○ Max water depth on crest (here h)

- If either of the two variables is too large → failure
- Pf is an OR probability.

Integral can be simple…



Based on Mares-Nasarre et al. (2024)
*Tolerability limits are purely illustrative.

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2. Concept of failure in reliability

We could include more variables: wave direction, wind speed…

Integrals are not always simple…

Example: crown wall on top of a breakwater under wave attack.

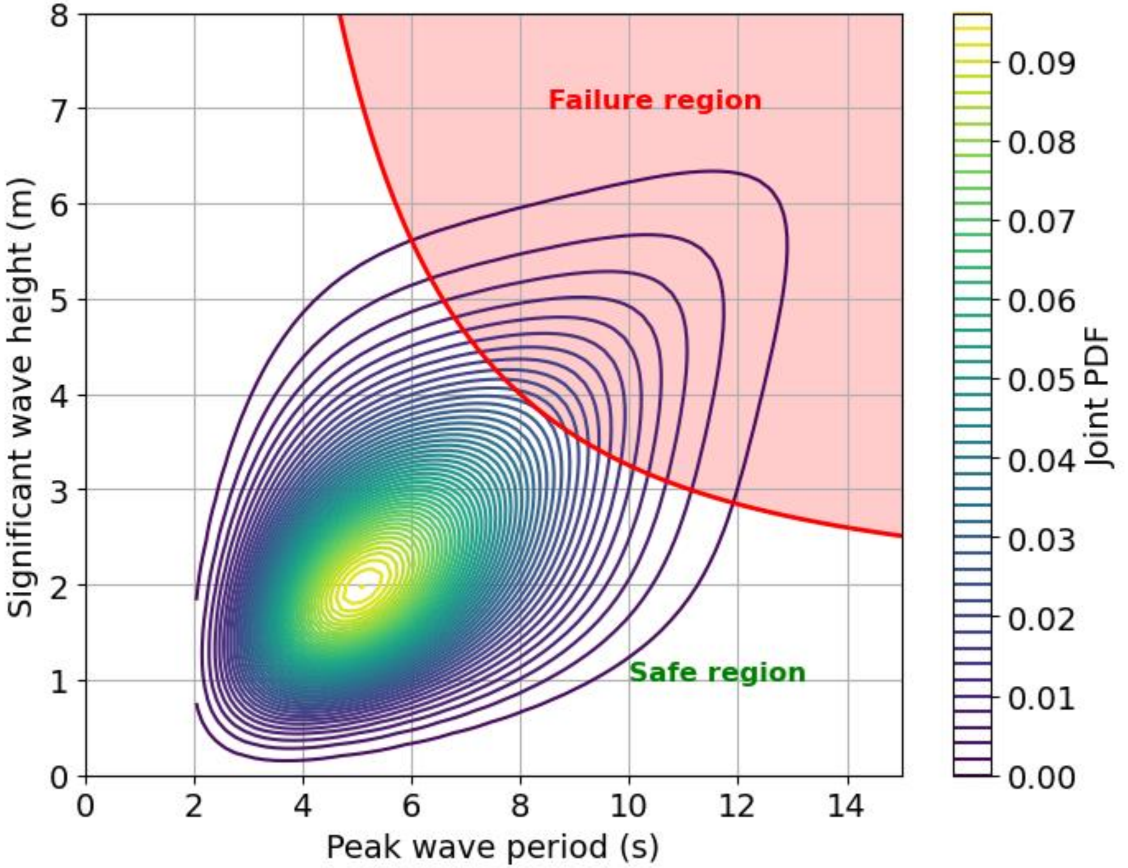
Loading (USACE, 2022): $F_h = \left(A_1 + A_2 \frac{H}{A_c}\right) \rho C_h L_{0p}$

Assuming a given geometry and wave length in deep waters:

$$F_h = 255.4HT^2 - 490.4T^2$$

Limit state:

$$g(x) = R - (255.4HT^2 - 490.4T^2)$$



In general: $P_f = \int_{g(\underline{X})<0} f_{\underline{X}}(\underline{x})\, d\underline{x}$

Outline

1. What is component reliability?
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3. Method 1: Monte Carlo
4. Method 2: FORM

Intermezzo

In general: $Pf = \int_{g(\underline{X})<0} f_{\underline{X}}(\underline{x})d\underline{x}$

In the past, reliability methods were divided in 5 groups:

- Level 0: deterministic.
- Level I (semi-probabilistic design): one characteristic value. E.g.: partial coefficients in codes.
- Level II (**approximation**): Variables are assumed normal-distributed → FORM (linear g(x)) → Analytical solution, can be solved by hand
- Level III (numerical): Use of joint distribution functions and computation of **actual probability of failure** without approximations → Monte Carlo → But now we have laptops!!
- Level IV (risk-based): risk used as a measure of reliability

Don't approximate if you have the tools

Based on Jonkman et al. (2017). Probabilistic Design: Risk and Reliability Analysis in Civil Engineering

Outline

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3. Monte Carlo

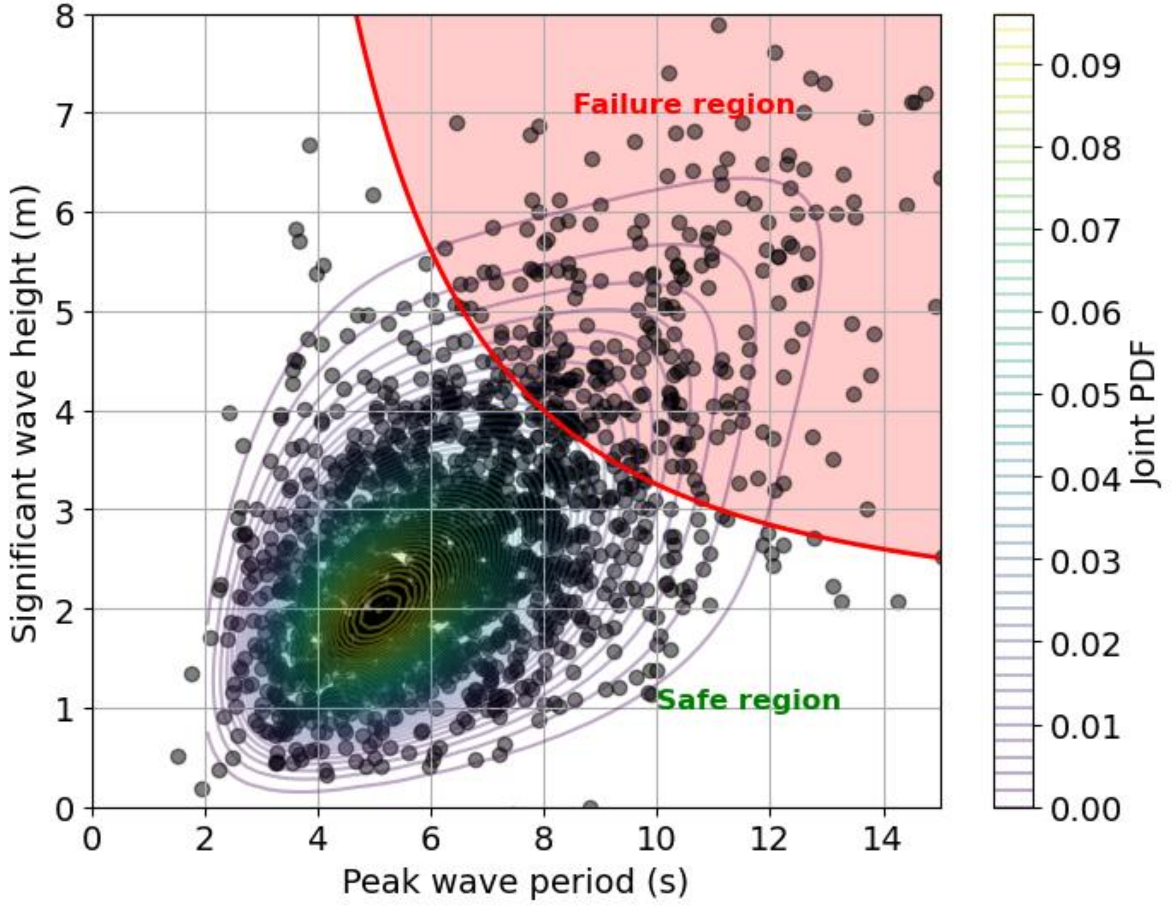
You have already used it… Does it ring a bell?

Solves the integral through simulation

$$\boldsymbol{Pf} = \int_{g(\underline{X})<0} \boldsymbol{f}_{\underline{X}}(\underline{\boldsymbol{x}})\boldsymbol{d}\underline{\boldsymbol{x}}$$

General outline:

1. For each random variable X_i, we generate *N* realizations using their joint distribution
2. For each set *j* = 1,…,N, we evaluate the limit state function $g(x_i)$
3. We count the number of sets where $g(x_i)<0$, N_f
4. $Pf = N_f/N$



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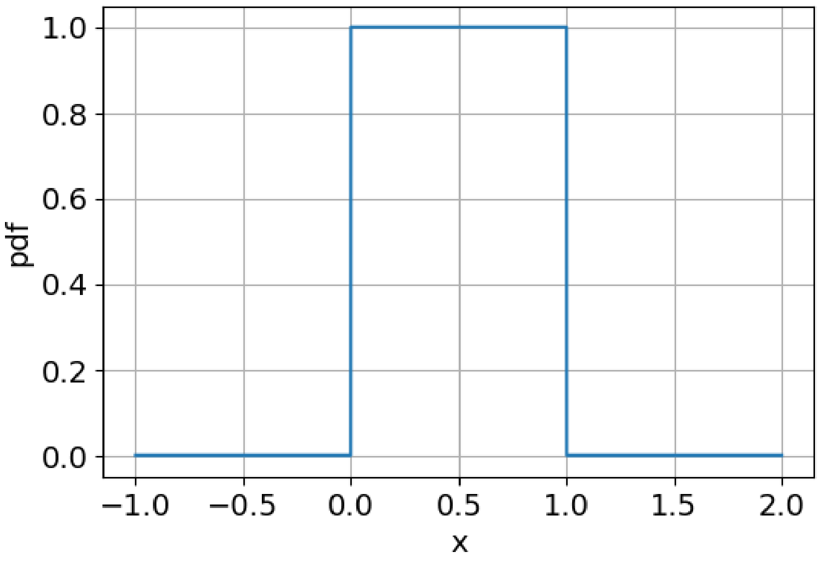
3. Monte Carlo – univariate sampling

Sampling one random variable, X: generate random values of X that follow its distribution

1. Draw a random value from a uniform distribution in [0,1], x_u
2. Let $X \sim F_X(x)$, $x = F_X^{-1}(x_u)$, where F_X^{-1} denotes the inverse CDF. We use x_u as the non-exceedance probability



3. Monte Carlo – inverse CDF

- **Cumulative distribution function (CDF), $F_X(x)$**
  - "Input": value of the random variable
  - "Output": non-exceedance probability
- **Inverse of the CDF, $F_X(x)^{-1}$**
  - "Input": non-exceedance probability
  - "Output": value of the random variable

Exponential distribution, known value of lambda

$$F_X(x) = 1 - e^{-\lambda x}$$

$$1 - F_X(x) = e^{-\lambda x}$$

$$-\ln(1 - F_X(x)) = \lambda x$$

$$\frac{-\ln\left(1 - F_X(x)\right)}{\lambda} = x$$

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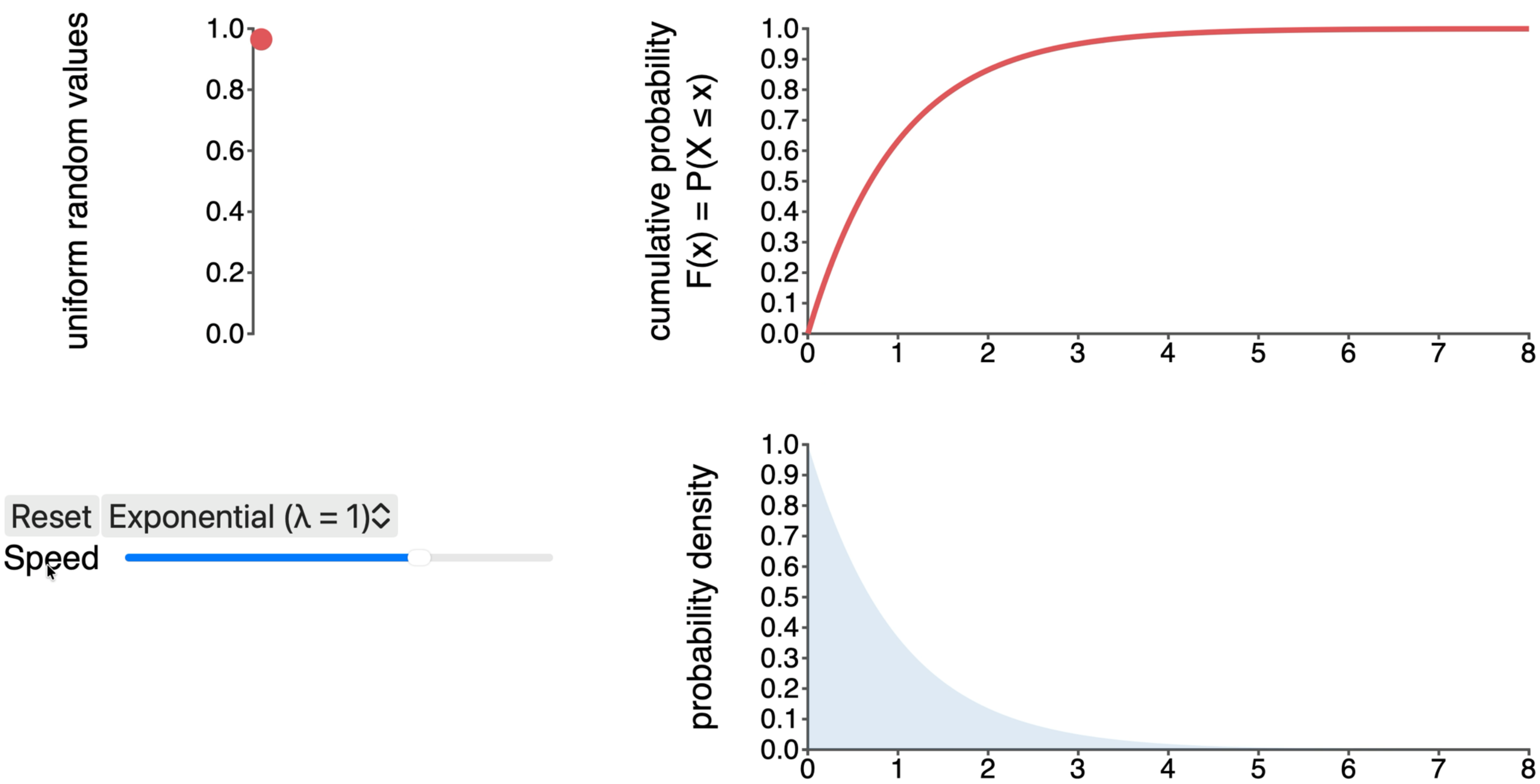
3. Monte Carlo



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Courtesy of Max Ramgraber

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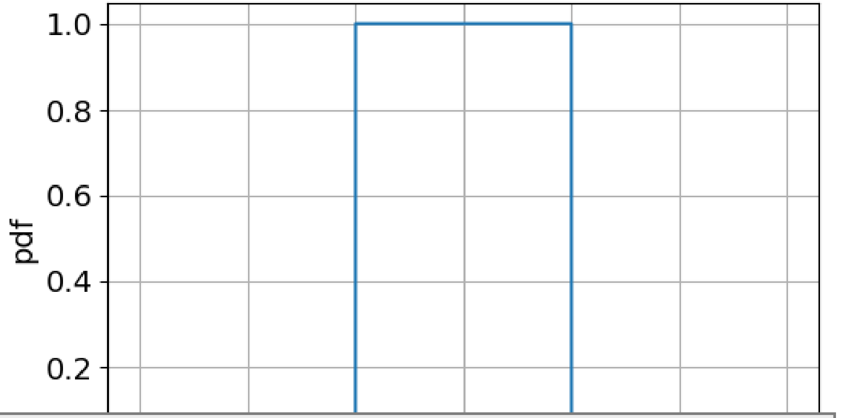


Small example: generate a sample for an Exponential distribution ($\lambda = 1$)

$x_u = 0.34$

$$x = \frac{-\ln(1-F_X(x))}{\lambda} = \frac{-\ln(1-0.34)}{1} = 0.416$$

From previous slide:

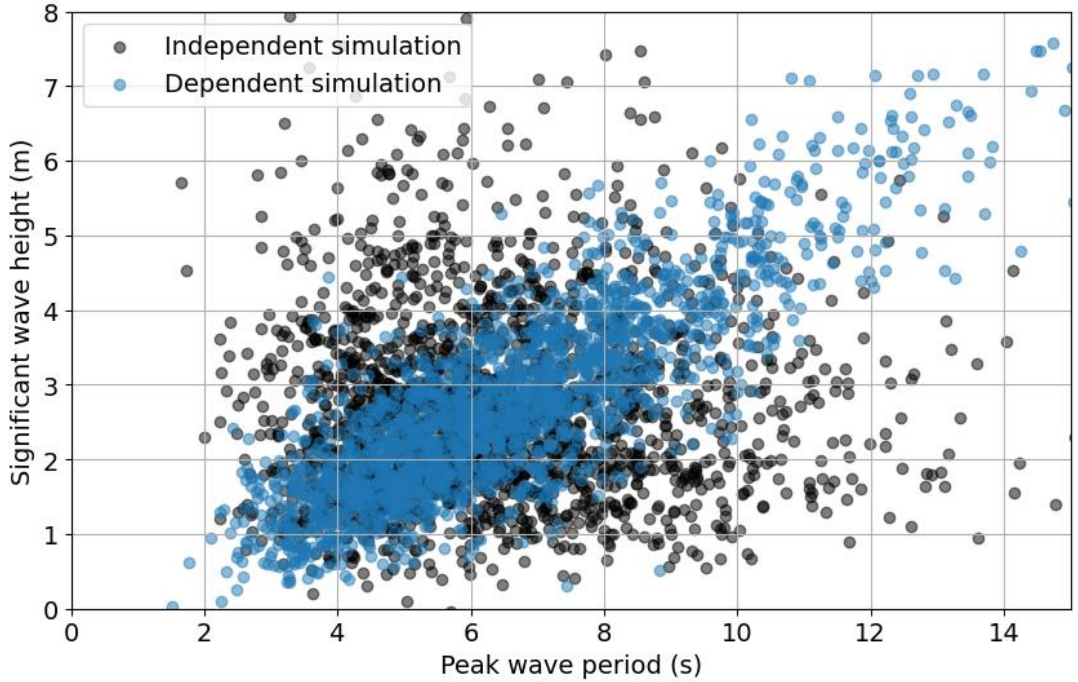
$$\frac{-\ln(1-F_X(x))}{\lambda} = x$$

3. Monte Carlo – more than one random variable

Two situations:

1. Variables are <u>independent</u>: sample independently each random variable (univariate sampling)

2. Variables are <u>NOT independent:</u> sample from a joint distribution to preserve the dependence between the generated samples



Does it matter to account for the dependence?

Toy example: assume that we approximate Pf = P[X>x AND Y>y]

Independence: Pf = 0.0025
Dependence: Pf = 0.046

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3. Monte Carlo

Solves the integral through simulation

$$Pf = \int_{g(\underline{X})<0} f_{\underline{X}}(\underline{x})\,d\underline{x}$$

General outline:

1. For each random variable X_i, we generate *N* realizations using their joint distribution
2. For each set *j* = 1,…,N, we evaluate the limit state function $g(x_i)$
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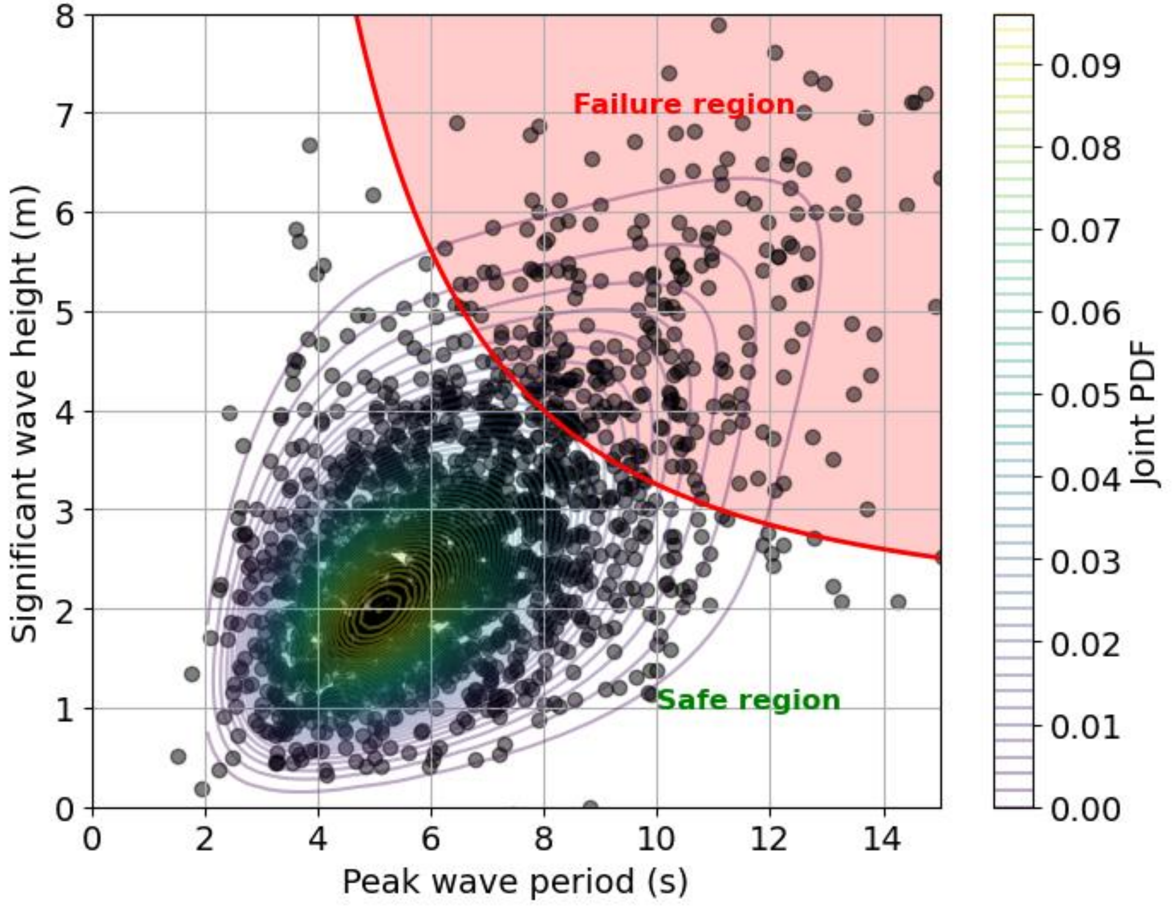

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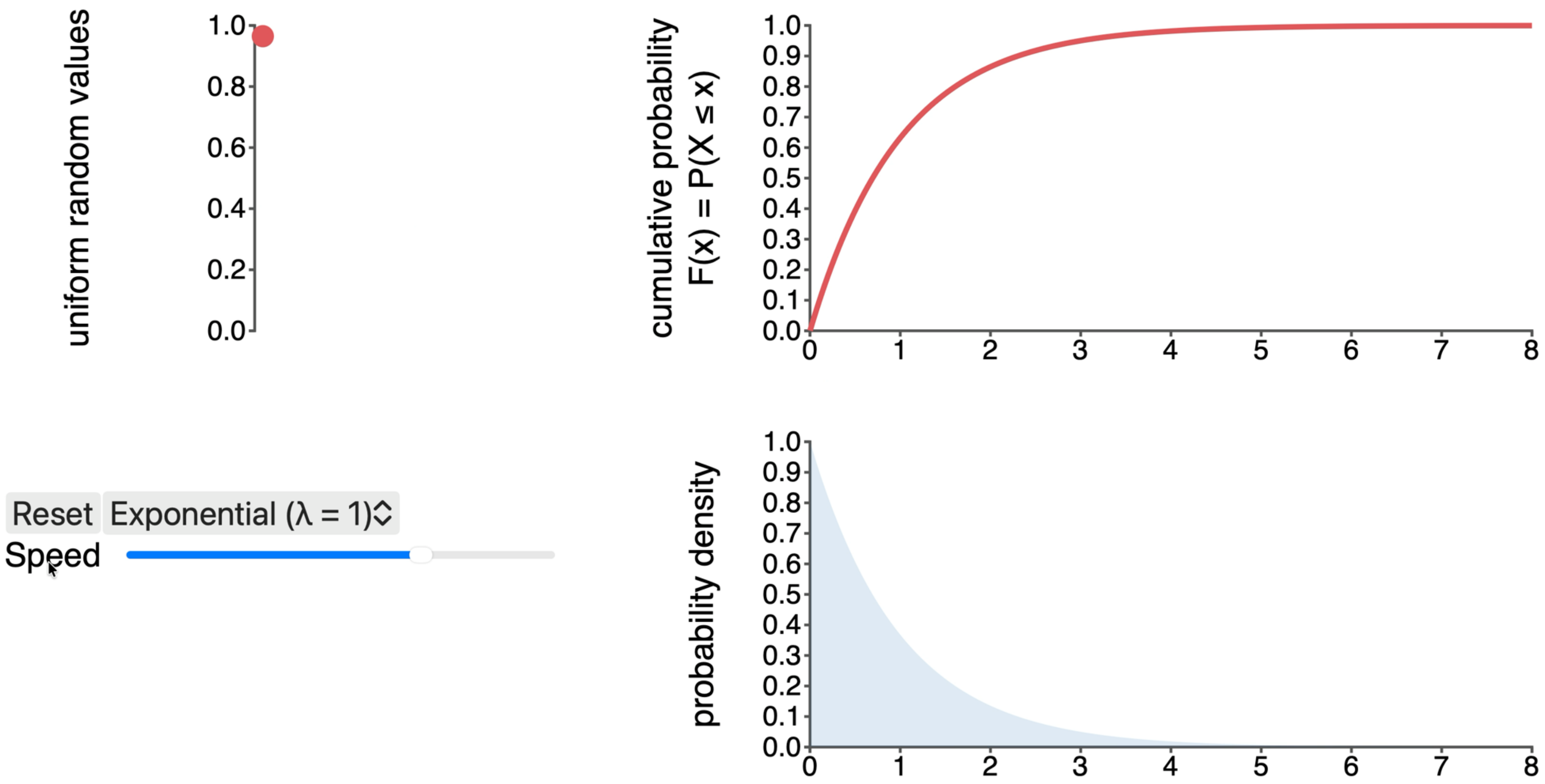
3. Monte Carlo – limitations/considerations



Courtesy of Max Ramgraber

Any hints about possible limitations or considerations when using Monte Carlo?

The higher the sample size, the better the approximation of Pf

↓

But the higher the computational cost

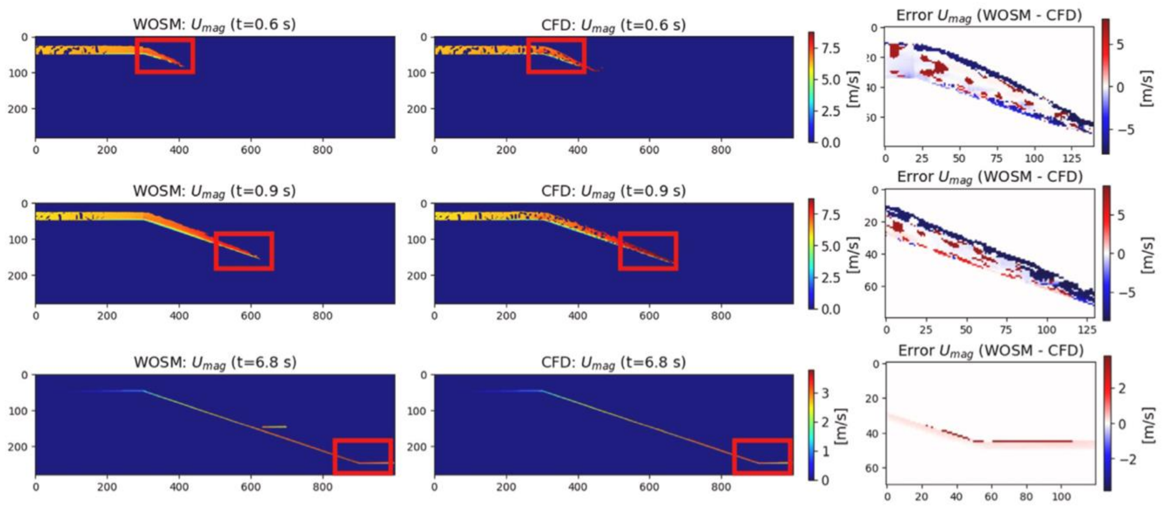
Small Pf: need for more samples, can get expensive

3. Monte Carlo – some ways of balancing the cost

Schrama et al. (2026)

Some proposals from literature:

- Use of <u>surrogate models</u> to substitute expensive runs of numerical models



Outline

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4. First Order Reliability Method (FORM)

Approximation method!

Assumptions:

- Each random variable X_i is or is transformable into independent standard normal variables
- Limit state function is approximated as linear in a design point and is continuous and differentiable
- Pf is dominated by one critical design point, denominated “most probable point”

A linear function of Gaussian variables follows a Gaussian distribution and can be represented by its mean and variance

It focuses on one single failure region and design point but there could be multiple regions and points. Choice may be arbitrary in some cases

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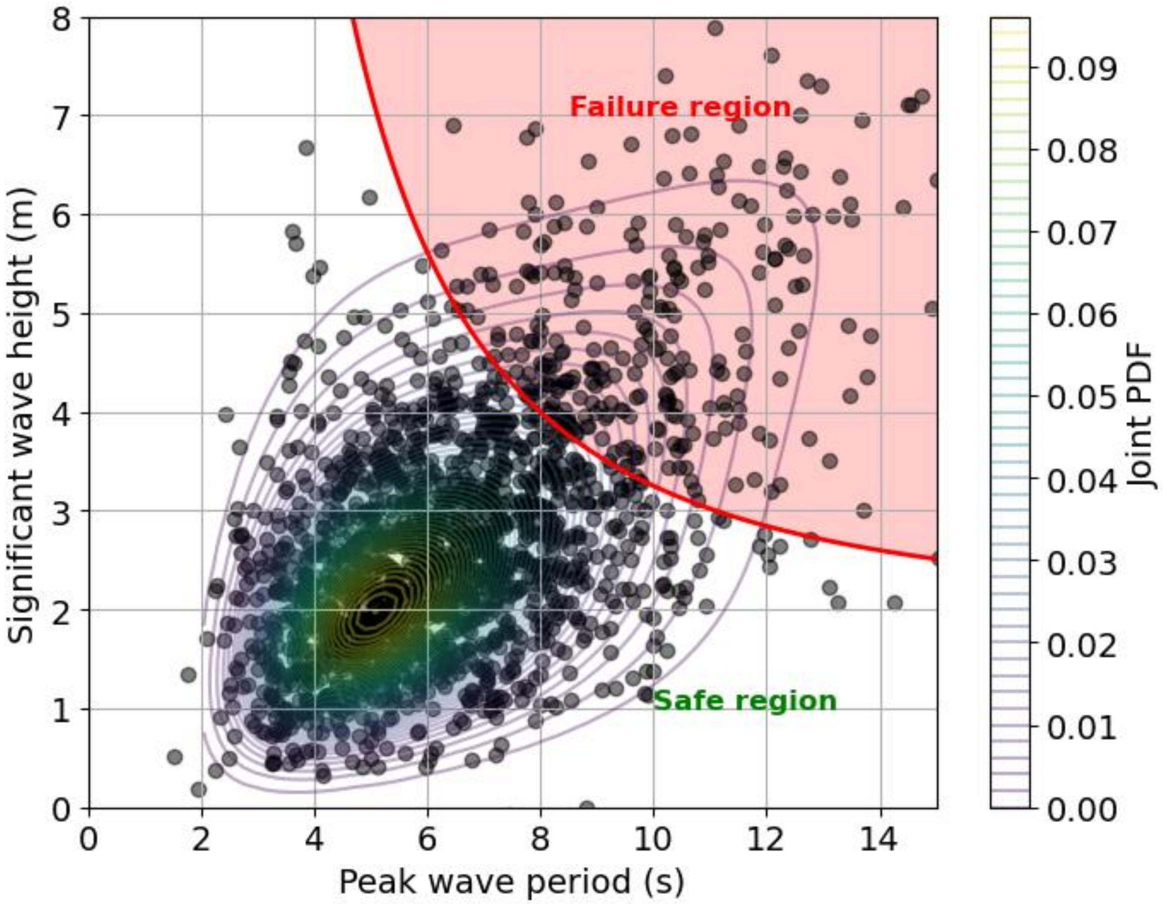
4. FORM – assumptions

When do its assumptions fail?

- Non-linear linear state functions or with multiple failure regions (e.g.: two failure modes)
- Discontinuous or non-differentiable limit state
- Highly non-normal or dependent variables (poor results of the transformation)
- Very small failure probabilities: tail becomes even more important!



4. FORM – example 1

$$E[X+Y] = E[X] + E[Y]$$

$$\sigma_{aX+bY} = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\mathrm{Cov}(X,Y)}$$

Linear g(x) and Gaussian variables

- $X_1 \sim N(2,1)$
- $X_2 \sim N(4,3)$
- $X_1 \perp X_2$
- $g(x) = 5 + X_1 - X_2$

$$\mu_G = 5 + 2 - 4 = 3$$

$$\sigma_G = \sqrt{1+3^2} = \sqrt{10} = 3.16$$

$P[g(x) < 0] = 0.17 \rightarrow$ Evaluate CDF of N(3, 3.16)

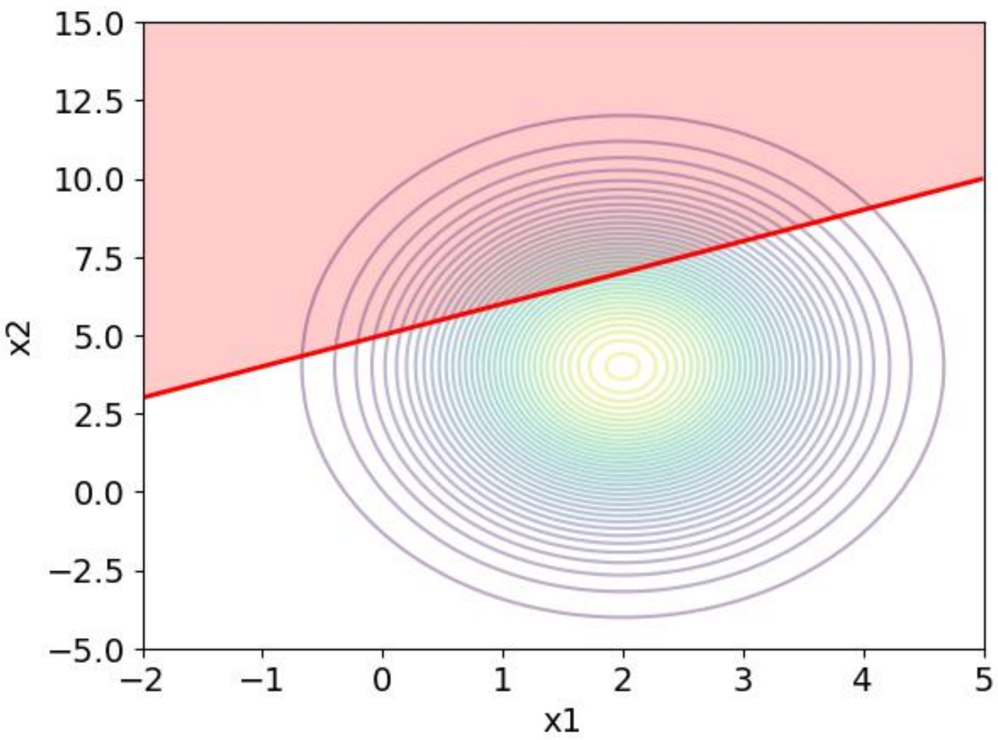
$$\beta = -\frac{0-3}{3.16} = +0.95$$

$$P(g(x) < 0) = \Phi\left(\frac{0-\mu_G}{\sigma_G}\right) = \Phi(-\beta)$$

Just uncertainty propagation!

g(x) is Gaussian and it is possible to compute its distribution.

Failure is defined as P[g(x)<0]



4. FORM – outline

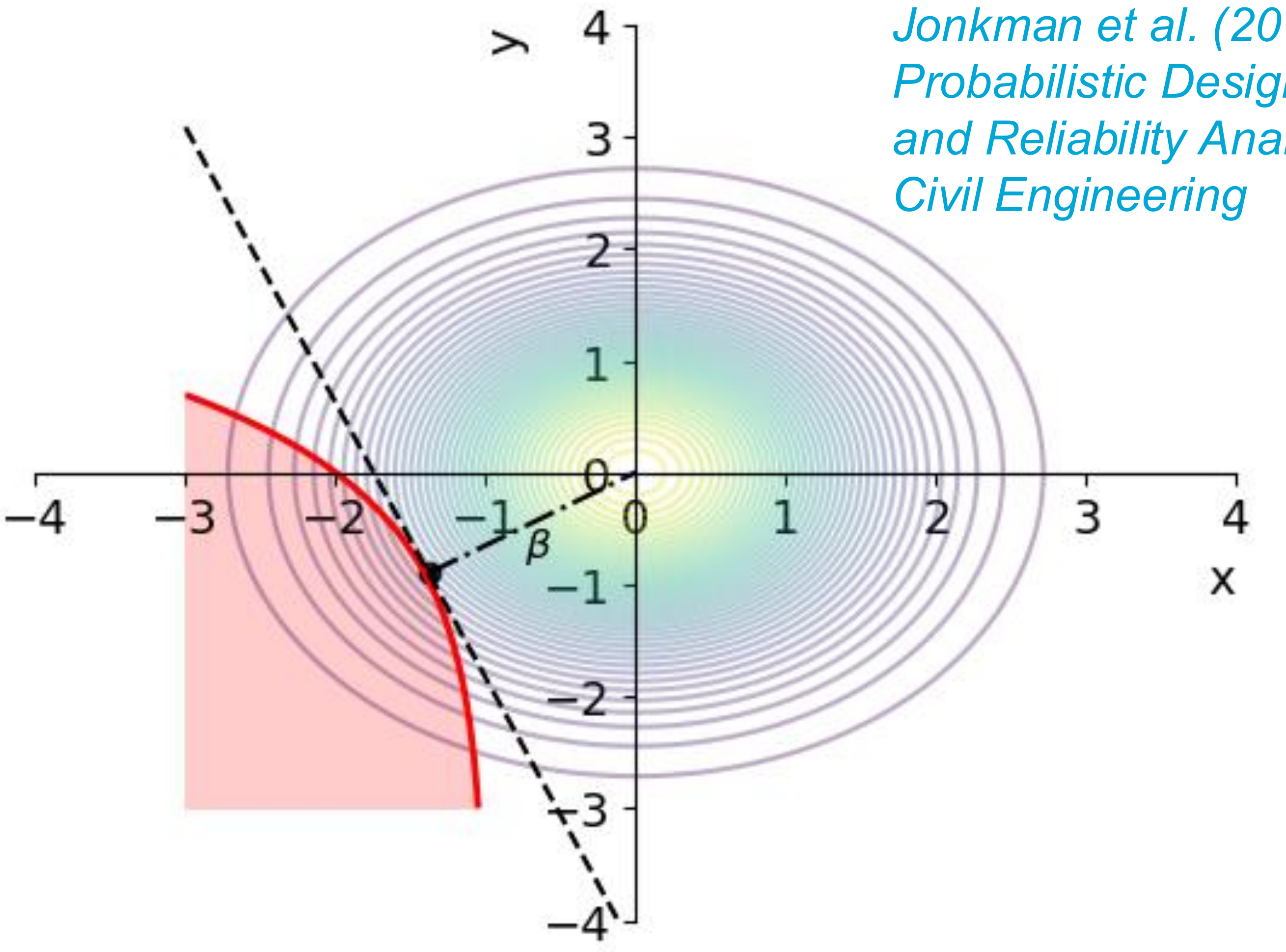
Outline: given a set of random variables X_i and a limit state $g(x_i)$

1. Transform X_i into independent standard normal variables
2. Determine the design point, x_d
3. Linearize $g(x)$ using 1st order Taylor series expansion
4. Compute the reliability index, β, and $Pf = \Phi(-\beta)$

We are not going to see the full derivation, only part of it.

For full derivation:

Jonkman et al. (2017). Probabilistic Design: Risk and Reliability Analysis in Civil Engineering



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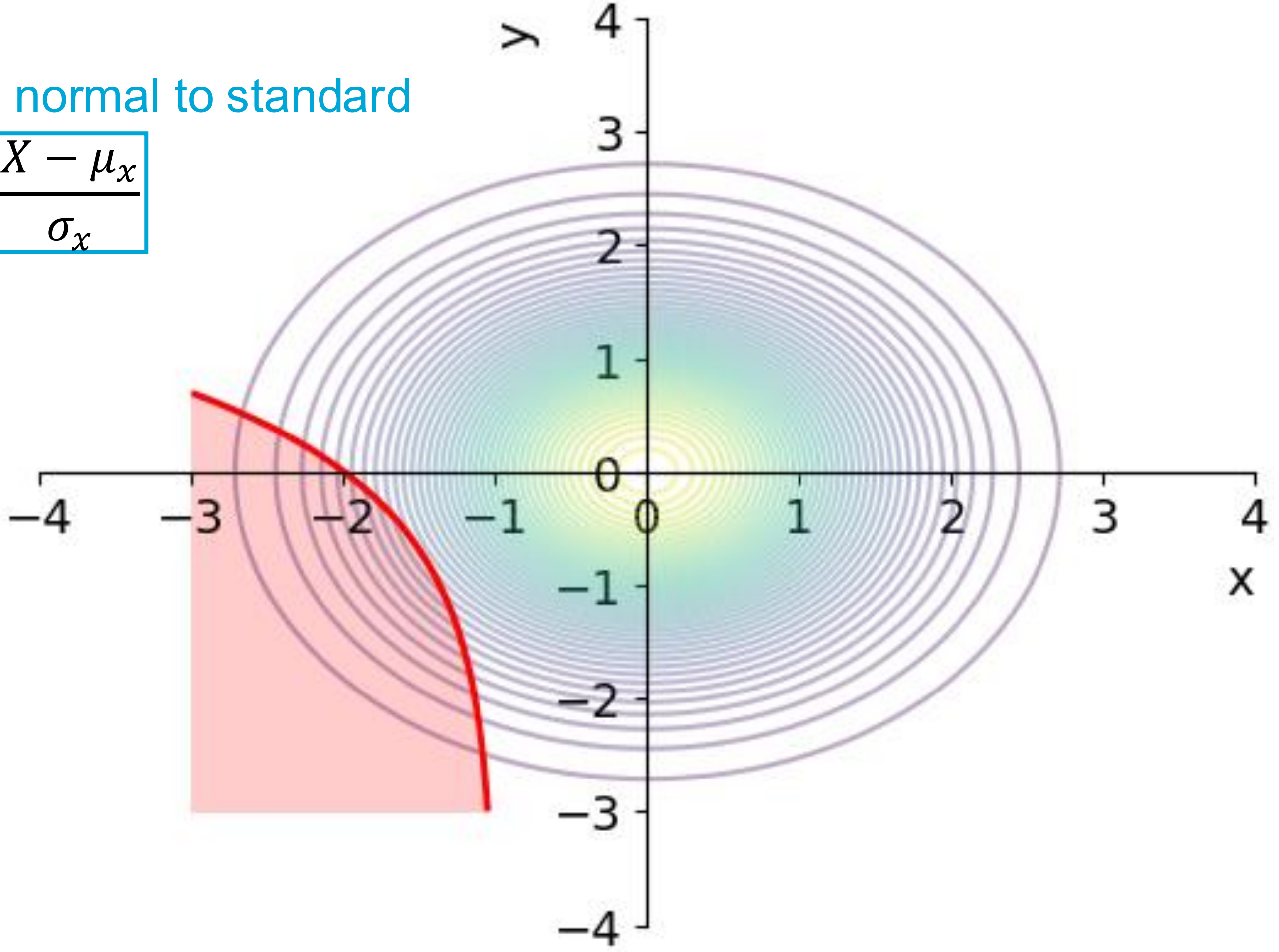
4. FORM – outline

Outline: given a set of random variables X_i and a limit state $g(x_i)$

1. Transform X_i into independent standard normal variables

From normal to standard

$$\boxed{U = \frac{X - \mu_x}{\sigma_x}}$$



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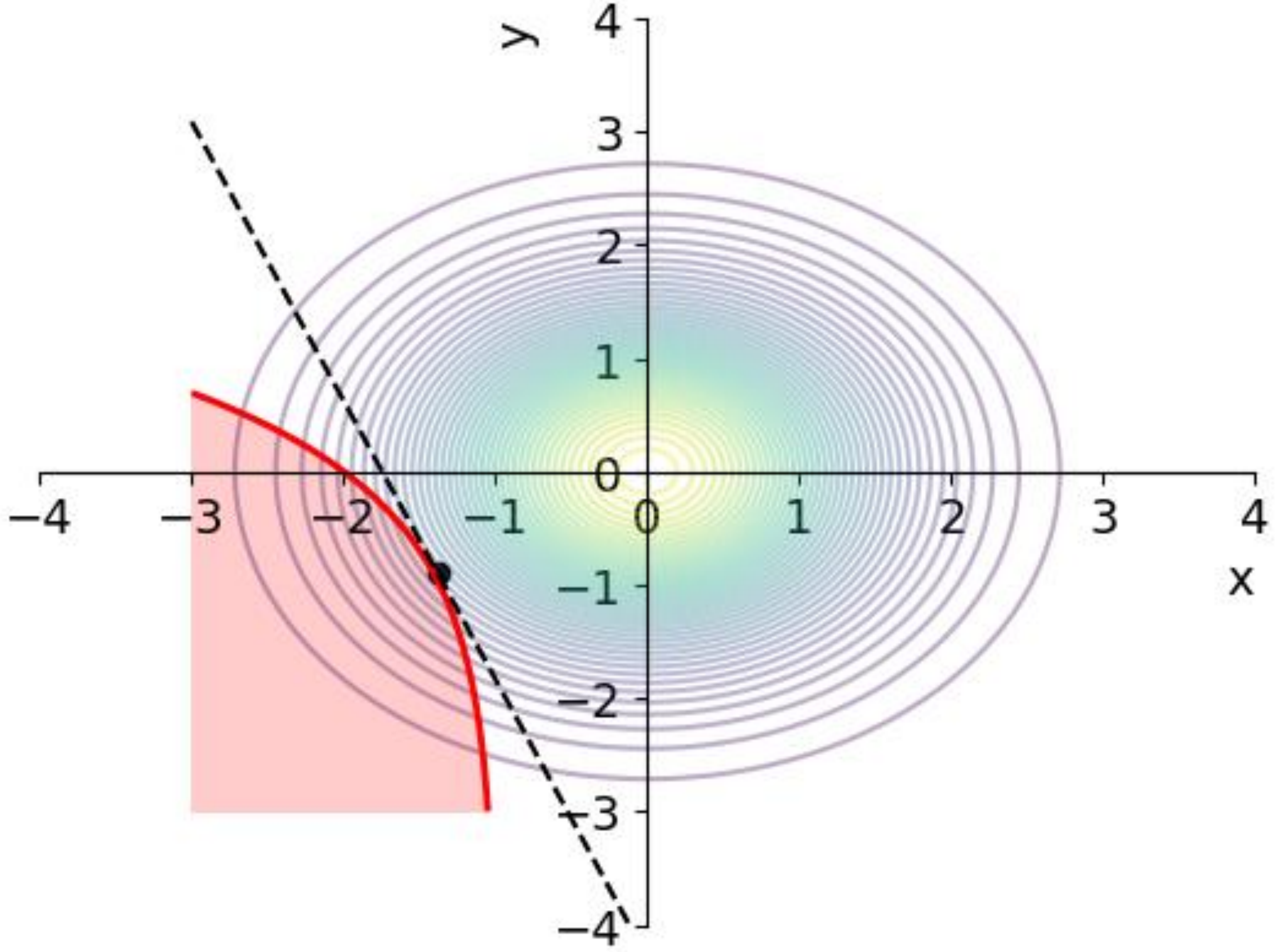

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$$\boxed{U = \frac{X - \mu_x}{\sigma_x}}$$

$$X = \sigma_x U + \mu_x$$

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1st order Taylor series expansion

$$g(\underline{U}) = g(\cancel{\underline{u}^*}) + \sum_{i=1}^{N} \frac{\partial g(\underline{u}^*)}{\partial U_i}(U_i - u_i^*) = \sum_{i=1}^{N} \frac{\partial g(\underline{u}^*)}{\partial U_i} U_i - \sum_{i=1}^{N} \frac{\partial g(\underline{u}^*)}{\partial U_i} u_i^*$$

The factor of the Hesse normal form:

If: $g(X) = a_0 + a_1 X_1 + a_2 X_2 = 0$

$$\beta = \frac{a_0}{\sqrt{a_1^2 + a_2^2}}$$

We can identify:

$$\left.\begin{aligned} a_0 &= -\sum_{i=1}^{N} \frac{\partial g(\underline{u}^*)}{\partial U_i} u_i^* \\ a_i &= \sum_{i=1}^{N} \frac{\partial g(\underline{u}^*)}{\partial U_i} \end{aligned}\right] \quad \beta = \frac{-\sum_{i=1}^{N} \frac{\partial g(\underline{u}^*)}{\partial U_i} u_i^*}{\sqrt{\sum_{i=1}^{N} \left[\frac{\partial g(\underline{u}^*)}{\partial U_i}\right]^2}} = \frac{-\sum_{i=1}^{N} \frac{\partial g(\underline{x}^*)}{\partial U_i} (x_i^* - \mu_i)}{\sqrt{\sum_{i=1}^{N} \left[\frac{\partial g(\underline{x}^*)}{\partial X_i} \sigma_i\right]^2}}$$

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$$\boxed{U = \frac{X - \mu_x}{\sigma_x}}$$

$$X = \sigma_x U + \mu_x$$

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Before: $a_i = \sum_{i=1}^{N} \frac{\partial g(\underline{u}^*)}{\partial U_i}$

$$\beta = \frac{-\sum_{i=1}^{N} \frac{\partial g(\underline{u}^*)}{\partial U_i} u_i^*}{\sqrt{\sum_{i=1}^{N} \left[\frac{\partial g(\underline{u}^*)}{\partial U_i}\right]^2}}$$

We can also define:

$$\alpha_i = \frac{a_i}{\sqrt{\sum_{j=1}^{n} (a_j)^2}} = \frac{\frac{\partial g(\underline{u}^*)}{\partial U_i}}{\sqrt{\sum_{j=1}^{n} \left(\frac{\partial g(\underline{u}^*)}{\partial U_j}\right)^2}} = \frac{\frac{\partial g(\underline{x}^*)}{\partial X_i}\sigma_i}{\sqrt{\sum_{j=1}^{n} \left(\frac{\partial g(\underline{x}^*)}{\partial X_i}\sigma_i\right)^2}}$$

Which are the **weight or sensitivity factors**. Indicate the relative importance of the standard deviation of a random variable to the reliability index.

So:

$$\beta = -\sum_{i=1}^{n} \alpha_i u_i^* \rightarrow u_i^* = -\alpha_i \beta$$

$$x_i^* = \mu_i - \alpha_i \beta \sigma_i$$

4. FORM – outline

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1. Transform X_i into independent standard normal variables
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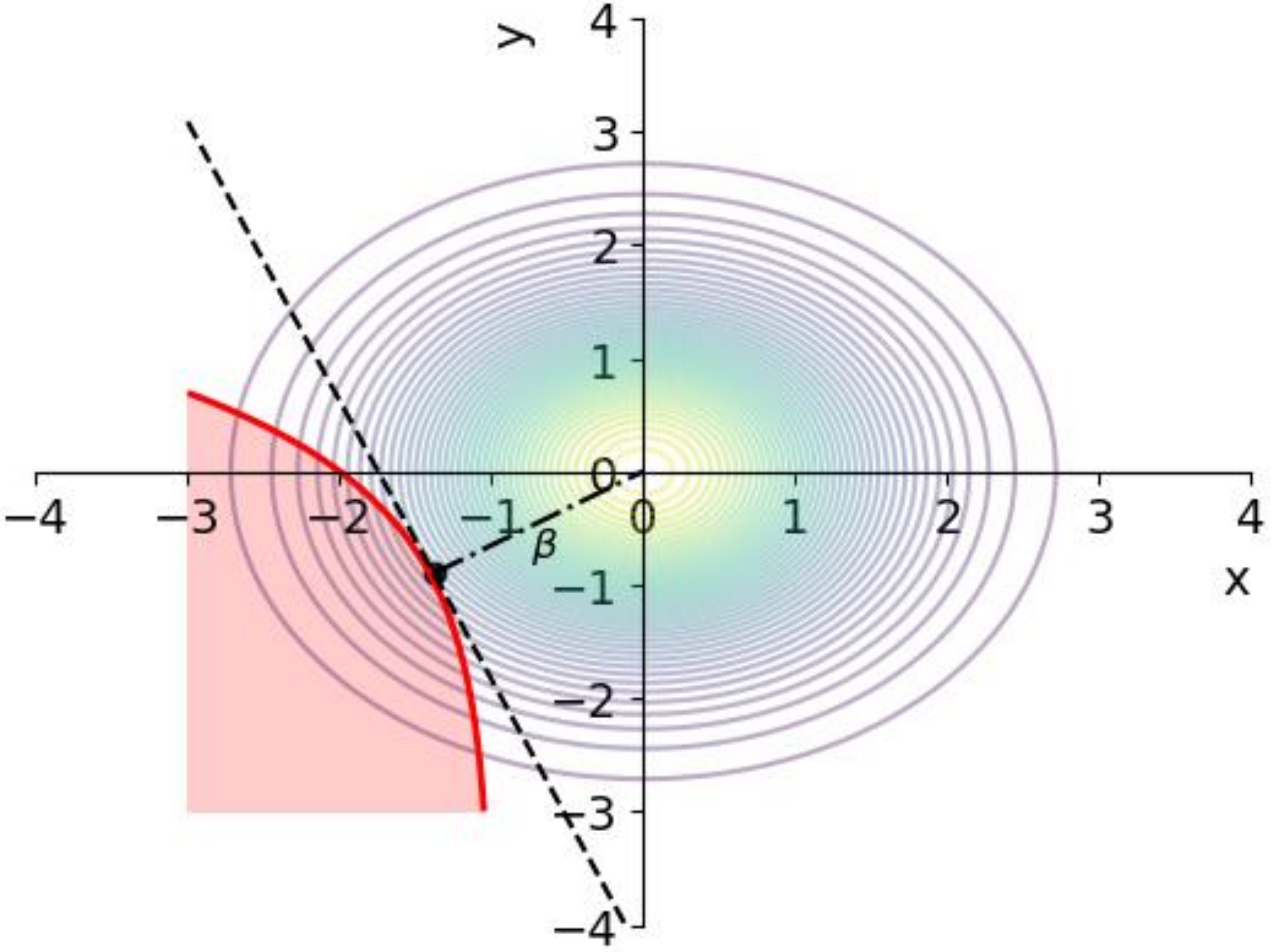
3. Linearize g(x) using 1st order Taylor series expansion
4. Compute the reliability index, β, and Pf = Φ(- β)



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4. Compute the reliability index, β, and $\boxed{Pf = \Phi(-\beta)}$

$$\alpha_i = \frac{\dfrac{\partial g(\underline{u}^*)}{\partial U_i}}{\sqrt{\sum_{j=1}^{n}\left(\dfrac{\partial g(\underline{u}^*)}{\partial U_j}\right)^2}} \qquad (1)$$

I need the design point!

$$u_i^* = -\alpha_i \beta \qquad (2)$$

$$x_i^* = \mu_i - \alpha_i \beta \sigma_i \qquad (3)$$

I define a system of equations using (1) and g(U) with (2) to solve for β and α_i (for each random variable) at a random point.

Using the obtained β and α_i, I recalculate the design point using (3).

I iterate until the solution converges.

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4. FORM – example 2

Non-Linear g(x) and Gaussian variables

- $X_1 \sim N(2, 1)$
- $X_2 \sim N(4, 3)$
- $X_1 \perp X_2$
- $g(x) = 5 + X_1^2 - X_2$

1 $U_1 = \dfrac{X_1 - 2}{1}$ $U_2 = \dfrac{X_2 - 4}{3}$

Rewrite g(<u>X</u>) as function of <u>U</u>:

$$g(\underline{U}) = 5 + (U_1 + 2)^2 - (3U_2 + 4) = 5 - U_1^2 + 4U_1 - 3U_2$$

2 **At the design point:** $u_i^* = -\alpha_i \beta$

$$g(\underline{U^*}) = 5 - \alpha_1^2\beta^2 - 4\alpha_1\beta + 3\alpha_2\beta = 0$$

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4. FORM – example 2

Non-Linear g(x) and Gaussian variables

- $X_1 \sim N(2,1)$
- $X_2 \sim N(4,3)$
- $X_1 \perp X_2$
- $g(x) = 5 + X_1^2 - X_2$

$$g(\underline{U}) = 5 - U_1^2 + 4U_1 - 3U_2$$

$$g(\underline{U}^*) = 5 - \alpha_1^2\beta^2 - 4\alpha_1\beta + 3\alpha_2\beta = 0 \qquad \beta = \frac{5}{\alpha_1^2\beta + 4\alpha_1 - 3\alpha_2}$$

3

$$\alpha_i = \frac{\frac{\partial g(\underline{u}^*)}{\partial U_i}}{\sqrt{\sum_{j=1}^{n}\left(\frac{\partial g(\underline{u}^*)}{\partial U_j}\right)^2}} \qquad u_i^* = -\alpha_i\beta$$

$$\frac{\partial g(\underline{u})}{\partial U_1} = -2U_1 + 4 \qquad \frac{\partial g(\underline{u})}{\partial U_2} = -3$$

$$\alpha_1 = \frac{-2u_1^* + 4}{\sqrt{(-2u_1^* + 4)^2 + (-3)^2}} = \frac{2\alpha_1\beta + 4}{\sqrt{(2\alpha_1\beta + 4)^2 + (-3)^2}}$$

$$\alpha_2 = \frac{-3}{\sqrt{(-2u_1^* + 4)^2 + (-3)^2}} = \frac{-3}{\sqrt{(2\alpha_1\beta + 4)^2 + (-3)^2}}$$

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4. FORM – example 2

Non-Linear g(x) and Gaussian variables

- $X_1 \sim N(2, 1)$
- $X_2 \sim N(4, 3)$
- $X_1 \perp X_2$
- $g(x) = 5 + X_1^2 - X_2$

$$x_i^* = \mu_i - \alpha_i \beta \sigma_i$$

$$x_1^* = 2 - 0.5\ 2\ 1 = 1$$

$$x_2^* = 4 - (-0.5)2\ 3 = 7$$

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$$\beta = \frac{5}{\alpha_1^2 \beta + 4\alpha_1 - 3\alpha_2}$$

$$\alpha_1 = \frac{2\alpha_1 \beta + 4}{\sqrt{(2\alpha_1\beta + 4)^2 + (-3)^2}}$$

$$\alpha_2 = \frac{-3}{\sqrt{(2\alpha_1\beta + 4)^2 + (-3)^2}}$$

$$x_i^* = \mu_i - \alpha_i \beta \sigma_i$$

Initial values:

β: evaluate in mean values

α_i: equal, positive for resistance, negative for load

4 **Choose an initial value and iterate:**

	Initial value	It 1	It 2	It 3	It 4
β	2				
α_1	0.5				
α_2	-0.5				
X*					

4. FORM – example 2

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$$x_i^* = \mu_i - \alpha_i\beta\sigma_i$$

Initial values:

β: evaluate in mean values

α_i: equal, positive for resistance, negative for load

4 **Choose an initial value and iterate:**

	Initial value	It 1	It 2	It 3	It 4
β	2	1.25			
α_1	0.5	0.89			
α_2	-0.5	-0.45			
X*	(1, 7)				

4. FORM – example 2

Non-Linear g(x) and Gaussian variables

- $X_1 \sim N(2, 1)$
- $X_2 \sim N(4, 3)$
- $X_1 \perp X_2$
- $g(x) = 5 + X_1^2 - X_2$

Pf = Φ(- β) = 0.19

Pf = 0.10 (Monte Carlo)

$$\beta = \frac{5}{\alpha_1^2\beta + 4\alpha_1 - 3\alpha_2}$$

$$\alpha_1 = \frac{2\alpha_1\beta + 4}{\sqrt{(2\alpha_1\beta + 4)^2 + (-3)^2}}$$

$$\alpha_2 = \frac{-3}{\sqrt{(2\alpha_1\beta + 4)^2 + (-3)^2}}$$

$$x_i^* = \mu_i - \alpha_i\beta\sigma_i$$

Initial values:

β: evaluate in mean values

α_i: equal, positive for resistance, negative for load

4 **Choose an initial value and iterate:**

	Initial value	It 1	It 2	It 3	It 4
β	2	1.25	0.85	0.87	0.89
α_1	0.5	0.89	0.90	0.88	0.88
α_2	-0.5	-0.45	-0.43	-0.48	-0.48
X*	(1, 7)	-	-	-	(1.22, 5.28)

4. FORM – what happens if…?

- **Non-Gaussian** variables → if CDF is known, transform to uniform and then to standard

- Variables are **dependent** → Rosemblatt transformation or better go to MORE!

- **Selection of design point** → different methods
  - Direct iteration over limit state function
  - <u>Transformation to normal variables</u>
  - Other optimization techniques such as Hasofer-Lind-Rackwitz-Fiessler (HL-RF) algorithm

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Intermezzo

In general: $\boldsymbol{Pf} = \int_{g(\underline{X})<0} \boldsymbol{f}_{\underline{X}}(\underline{x})\, d\underline{x}$

In the past, reliability methods were divided in 5 groups:

- Level 0: deterministic.
- Level I (semi-probabilistic design): one characteristic value. E.g.: partial coefficients in codes.
- Level II (**approximation**): Variables are assumed normal-distributed → FORM (linear g(x)) → Analytical solution, can be solved by hand
- Level III (numerical): Use of joint distribution functions and computation of **actual probability of failure** without approximations → Monte Carlo → But now we have laptops!!
- Level IV (risk-based): risk used as a measure of reliability

Don't approximate if you have the tools

Based on Jonkman et al. (2017). Probabilistic Design: Risk and Reliability Analysis in Civil Engineering

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Outline

1. What is component reliability?
2. Concept of failure in reliability
3. Method 1: Monte Carlo
4. Method 2: FORM

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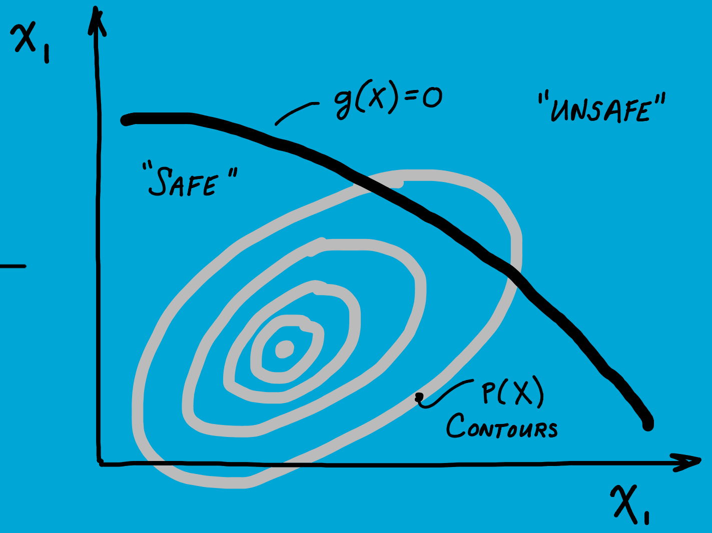
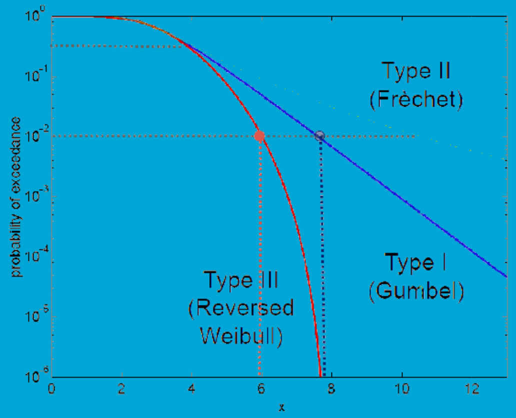
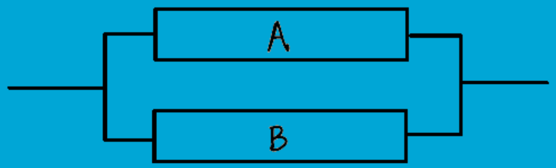
CIEM42X0 Probabilistic Design

Hydraulic and Offshore Structures (HOS) Track

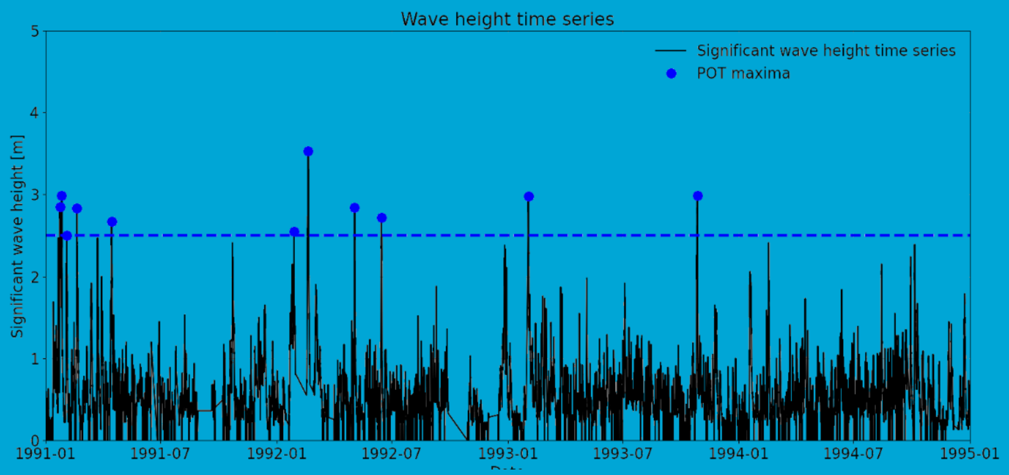
Civil Engineering MSc Program







Questions?



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